SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I sathish-M/192424 358 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

M. Sathish

Signature of the student:

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

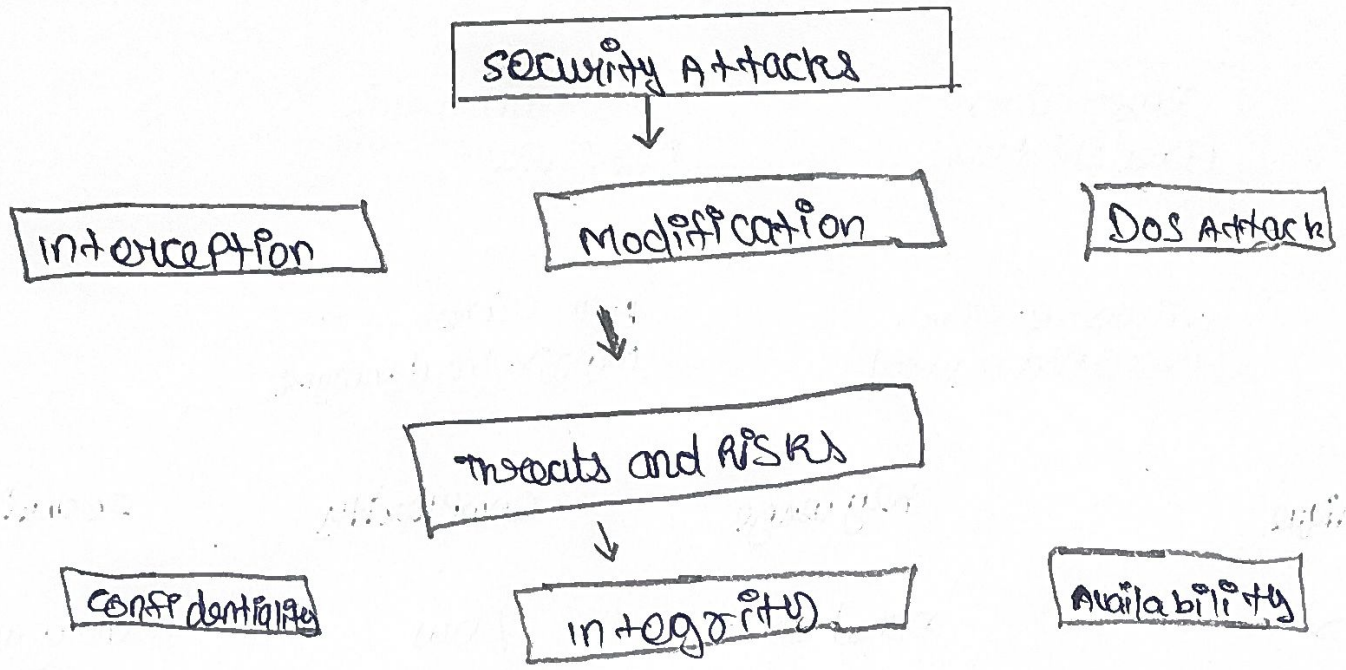
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Concept mapping

Creates a concept map linking the following: security Attacks, Threats, Risks and security Goals (Confidentiality, Integrity, Availability, Perform the following tasks.

Show how each attack leads to specific threats and risks.



Attack	threat or risk	Real-world example
Interception	unauthorized disclosure of information	Stealing passwords over unsecured wifi
Modification	unauthorized alteration of information	Changing online transaction details.
DOS attack	service interruption	Making a website unavailable through excessive malicious traffic.

b) Security goals and mitigation.

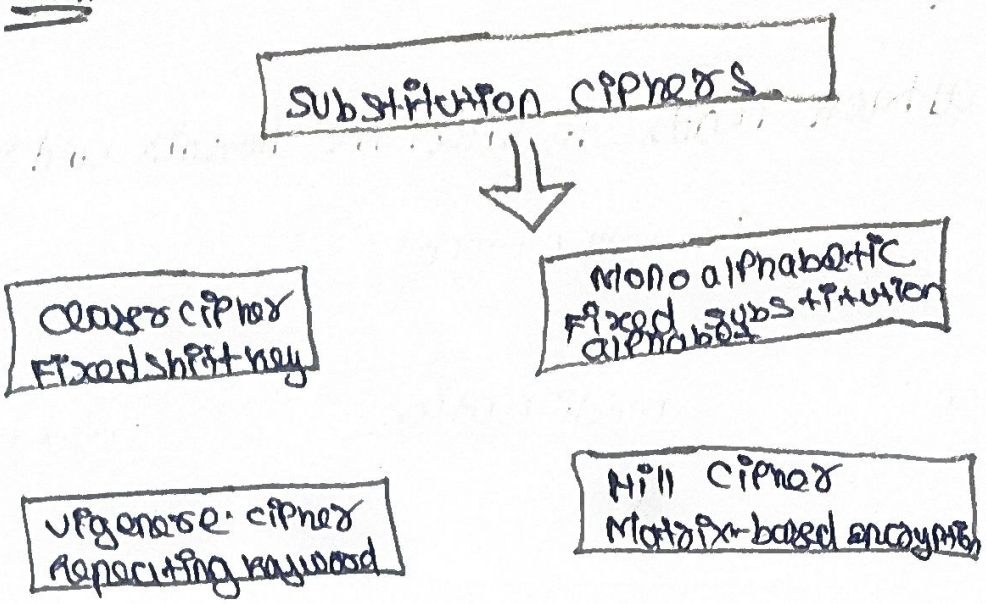
* Confidentiality: Encryption prevents unauthorized access to sensitive information.

* Integrity: Hashes and digital signatures help detect unauthorized changes.

* Availability: Backups, redundant systems and traffic filtering help maintain service availability.

2) Comparison of Substitution techniques:

Concept map:-

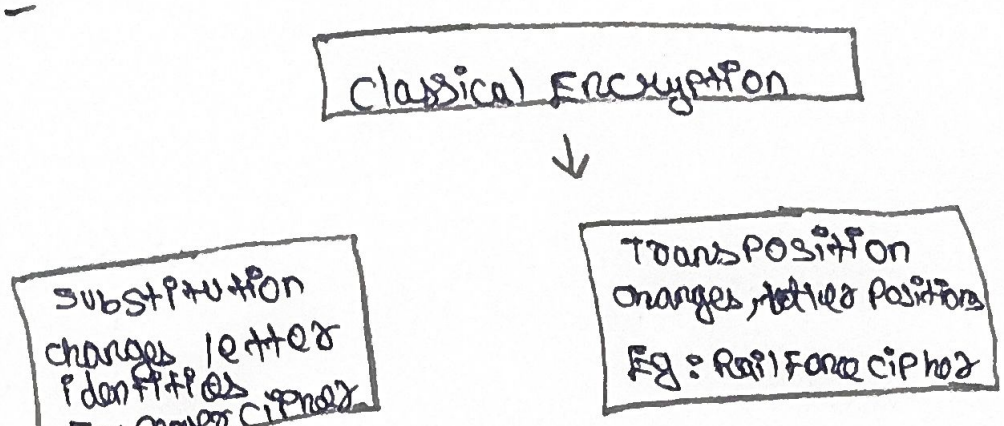


technique	key usage	Complexity	security
Ceasers	Fixed shift	Low	very weak
Monoalphabetic	Substitution alphabet	Moderate	vulnerable to frequency analysis
Vigenere	Repeating keyword	Moderate	vulnerable to repeating key analysis
Hill	invertible matrix	Higher	Vulnerable to known plaintext attacks.

Analysis: The Hill cipher hides individual-letter frequency patterns by encrypting blocks of letters. However, it is not completely secure. All four classical techniques are unsuitable for modern confidential communication.

Classical Encryption schemes.

concept map :-



Differences and similarities:-

Feature	Substitution	Transposition
operation	Replaces letters	Rearranges letters
Letters identities	changed	Preserved
Letters order	Generally preserved	Changed
Example	ceaser cipher	Rail Fence cipher

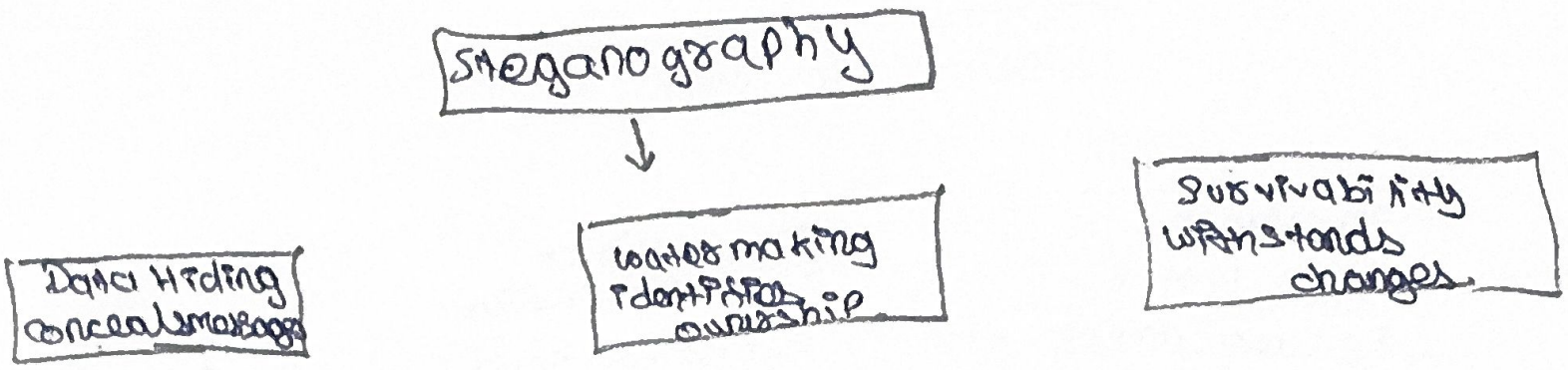
b) Effect on data structures and security.

* Substitution : Changes the letters but may preserve frequency patterns.
* Transposition : Rearranges letters but preserves their frequencies

c) Which is more vulnerable?

simple substitution ciphers are vulnerable to frequency analysis while transposition ciphers are vulnerable because the original letters remain unchanged. Both provide weak security compared with modern encryption.

4) Steganography = Data Hiding, watermarking and survivability.



a) Relationships :-

* Data hiding : Embeds confidential information within an image audio file or video.

* watermaking : Embeds ownership or authentication information into digital content.

* Survivability : Help embedded information remain recoverable after changes such as compression.

b) Contribution to security :-

Data hiding conceals communication, watermaking supports ownership verification and survivability helps embedded information remain useful after processing.

c) Real-world applications :-

Data hiding	concealing a message inside a image.
Watermaking	Protecting digital photographs with copy rights marks.
Survivability	Retaining a video watermark after compression.